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Piano Note Octave Decoding with Band Filtering

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Background

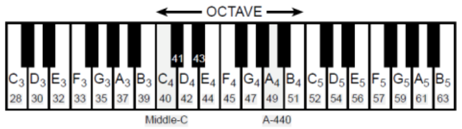
Standard Pianos have 88 keys, 7 octaves. Across each octave, there are 12 keys each with a unique frequency.

In this experiment, a method for decoding sinusoidal signals to a piano octave was established.

The principles used in these experiments are important in more than just music note decoding. They are also vital in operations related to telecommunication.

Introduction

Piano Keyboard anatomy showing octaves 3-5. Reference frequency of 440 Hz is found at note A4.



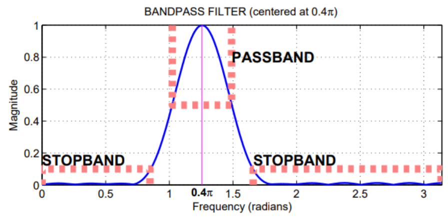
Methods

Simple bandpass filter equation (Rectangular Window):

$$h[n] = \frac{2}{L}\cos(\hat{\omega}_c n)\,,\, 0 \le n < L$$

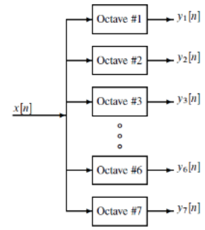
Bandpass filter bank equation (Hamming Window):

$$h[n] = (0.54 - 0.46\cos(2\pi n/(L-1)))\cos(\hat{\omega}_c(n-(L-1)/2)),$$
$$for\ n = 0,1,2,\ldots,L-1$$



Experiments

Design Passband Filter bank for octaves 2-6. Each Passband Filter is a unique length as a result of increased bandwidth Per octave. Bandpass filters only allow frequencies inside The passband through. This is ideal for note decoding.



Results

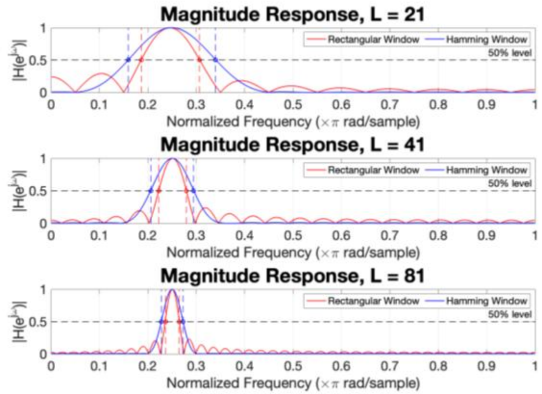
Each of the following demonstrate the results obtained through experimentation of Z-Transforms, Bandpass Filters, and Piano note/octave decoding.

Testing Windowing & Filtering Length

Windowing and Filter Length vs Passband Width (table)

Magnitude Response (plot)

L	Rectangular Window Passband Width (ω/π)	Hamming Window Passband Width (ω/π)
21	0.120	0.181
41	0.057	0.088
81	0.028	0.045

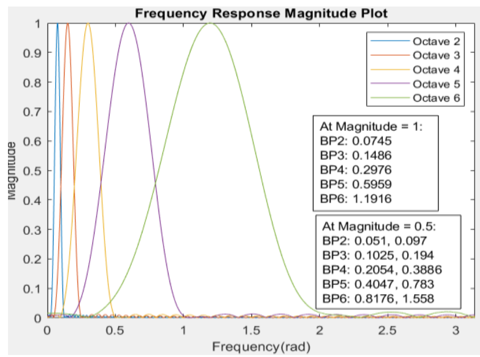


Bandpass Filter Bank Design

Calculated Frequencies (table)

Filter Frequency response (plot)

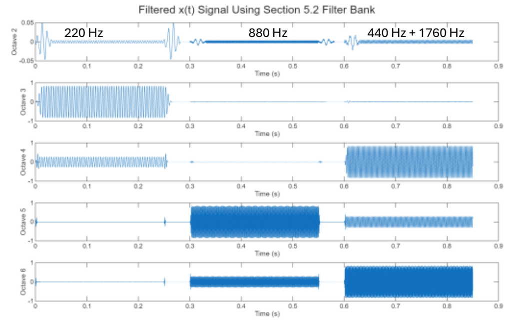
OCTAVE	Lower ω (rad/s)	Upper ω (rad/s)	Center ω (rad/s)
2	0.05137	0.09697	0.07417
3	0.1027	0.1939	0.1483
4	0.2055	0.3879	0.2967
5	0.411	0.7758	0.5934
6	0.8219	1.5516	1.1868



Piano Octave Decoding

Filtered unknown piano notes (plot)

Sampling frequency $f_s = 8000\ Hz$



Summary

This study focuses on creating simple & complex bandpass filters that can be applied to real world situations. In the case of this experiment, a collection of piano notes are processed through a bandpass filter bank to identify which piano note resides in one of the 1-7 octaves.

References

Lab P-14: Octave Band Filtering

Acknowledgments

Professor Neda Nategh

Ritaja Das

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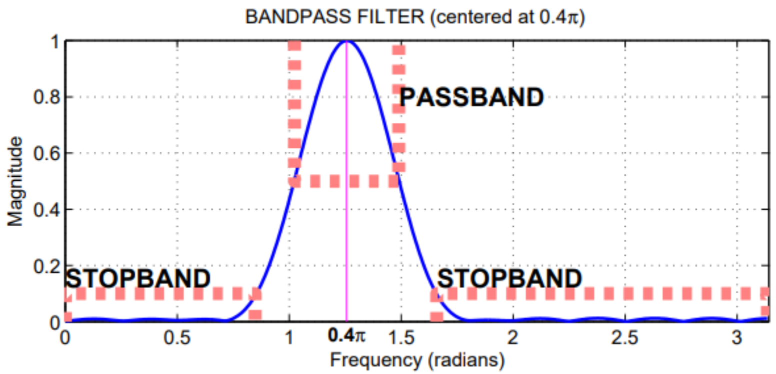

Summary

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